



Real-Time Revolutionizing Internal Defect Detection in Power Transformers by Leveraging Wavelet Transform and Deep Learning LSTM in Cascading Application

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Abstract

In recent times, the real-time and online identification and prediction of power transformer malfunctions play a vital part in maintaining the dependability and stability of electrical system operations. Early detection of transformer malfunctions is imperative to prevent unexpected outages and potential blackouts. Not only do such occurrences result in financial losses and physical damage, but they also pose a significant threat to the overall stability and functionality of the electrical grid. Therefore, this paper introduces an innovative approach leveraging long short-term memory techniques, coupled with the features of wavelet transformation, to effectively identify and predict internal defects (IDs) in power transformers amidst inrush currents (ICs) in online applications. A key highlight of this approach is its ability to accurately differentiate IDs from ICs within a quarter cycle of the fundamental frequency, enabling real-time fault prediction—a capability overlooked in previous research endeavors. Furthermore, a novel technique is proposed to mitigate the adverse impacts of current transformer saturation on the functionality of differential relays. The efficacy of the proposed methodology and the extracted features for deep learning applications is corroborated through a series of MATLAB simulations conducted on a three-phase transformer (33/11 kV, 31.5 MVA). Ultimately, the simulation results and testing validate the remarkable accuracy of 99.65% achieved by the proposed approaches in real-time fault detection for power transformers. This signifies a significant advancement in enhancing the reliability and operational efficiency of power systems.

Keywords Power transformer · Real-time internal fault detection · LSTM network · Wavelet transform

Abbreviations

LSTM	Long short-term memory
IC	Inrush current
IF	Internal fault
ID	Internal defect
PT	Power transformer
EKF	Extended kalman filter
CNN	Convolutional neural networks
BIGRU	Bidirectional gated recurrent unit
PNN	Probabilistic neural network
CT	Current transformer
EV	Energy value
THD	Total harmonic distortion

RT-BSWT	Real-time boundary stationary wavelet transform
SVM	Support vector machine
EWT	Empirical wavelet transform
MRA	Multi-resolution analysis
MMFE	Multi-scale multivariate fuzzy entropy
WT	Wavelet transform
IOT	Internet of things
PSO	Particle swarm optimization
DWT	Discrete wavelet transform

List of Symbols

N_w	Number of winding turns on the no-load side of the transformer
α	Represents the switching angle
$\Phi_{j,k}(t)$	Scaling function or mother wavelet
D_1	Signifies the detailed coefficients
X_t	The input of vectors of LSTM
O_t	Output gate activation vector at time t
W_f, W_i, W_o	The relevant gates' input kernels
b_f, b_i, b_o	The biases

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Y	Output labels indicating the fault condition
σ	The sigmoid function of logistics
θ	Parameters (weights and biases) of the LSTM network
φ	Magnetic flux within the iron core
U_m	Peak value of the supply voltage
A_0, A_1	Aproximate coefficients
f_t	Forget gate activation vector at time t
i_t	Input gate the activation vector at time t
c_t	The memory cell activation vector at time t
h_t	The activation vector of the hidden state at time t
U_f, U_i, U_o	The corresponding gates' recurrent kernels
$\tanh$	The activation function of hyperbolic tangent
$\otimes$	The product of matrix
CT	Current transformer

1 Introduction

1.1 Importance and Motivations

Since ancient times, the primary method of safeguarding power transformers (PTs) has typically been through the implementation of differential protection, yet the potential for malfunctioning during magnetizing instances remains a significant concern. PT internal defect currents and inrush currents (ICs) often present similar characteristics, thereby posing a challenge for differential protection algorithms. For instance, the simultaneous activation of two transformers in parallel may induce an abrupt change in magnetizing voltage, resulting in an inrush current. Similarly, the restoration of voltage following the removal of an external fault may trigger a substantial IC. The ability to differentiate between ICs and internal defects (IDs) presents a considerable challenge for differential protection systems, underscoring the critical need for real-time inner fault detection by power system operators and managers. PTs with capacities exceeding 10 MVA commonly rely on differential protection, and their failure can disrupt regular operation and power continuity [1]. While traditional fault diagnostic procedures for power transformers are frequently employed, they are often time-consuming and difficult to execute under typical working conditions for transformers. As a result, periodic inspection and post-fault diagnosis are commonly employed practices. Failure to accurately diagnose between ICs and IDs can eventually lead to transformer damage, impairing system performance, threatening worker safety, and incurring significant financial losses. IDs are accountable for almost 70% to 80% of power transformer damages, thereby posing a significant risk [2]. Addressing this challenge, differential protection systems must swiftly and accurately recognize IDs and promptly disconnect related circuit breakers.

However, PTs generate a substantial current (referred to as IC) upon energization, causing the windings to become magnetized. The problem arises from ICs, which are sudden spikes in current that occur during startup. The size of the transformer, any magnetic flux that may still be present in the core, the properties of the core, and the voltage level when the transformer is turned on are some of the variables that affect these currents. Often quite large ICs can interfere with protective systems meant to identify anomalies from operating as intended.

1.2 Literature Review

Recent advancements in transformer differential protection methods have been extensively explored in modern research literature. Among these approaches, harmonic-current restraint stands out as a well-established technique for discerning between IC and ID scenarios. Notably, it leverages the 2nd harmonic component, which typically ranges between 30 and 60% of the fundamental component [3]. This approach, as detailed in [2], utilizes the 2nd harmonic to prevent the erroneous operation of the differential relay during transformer energization and the occurrence of IC. The principle behind this method involves blocking the relay if the proportion of the 2nd harmonic to the primary harmonic component surpasses a predefined threshold, typically set at 15%. To enhance the reliability of the differential protection relay, researchers have explored incorporating additional harmonic components. In [4] this technique, the 2nd harmonic component is used to avoid incorrect work of the differential relay when energize the transformer and pass of IC. According to this method, if the 2nd harmonic's proportion to the primary harmonic component is greater than the threshold (typically 15%), the relay is blocked. For instance, in [5], the sum of the 2nd and 5th harmonics was considered. However, while total harmonic distortion (THD), as discussed in [6], shows promise in distinguishing between IC and ID scenarios based on higher THD values during inrush conditions, its application to a differential relay may lead to incorrect blocking or delayed operation. To address these challenges, novel methodologies have been proposed. For instance, In order to distinguish between IC and ID, a relationship between the differential current's energy coefficients is proposed in [7] using the real-time boundary stationary wavelet transform (RT-BSWT). However, this approach's inherent computational complexity and dependence on threshold values could lead to measurement errors. Other methods have also been investigated, such as those that use machine learning algorithms. In [8], a method utilizing support vector machine (SVM) and empirical wavelet transform (EWT) is presented for event identification in power transformers. However, the lengthy data training process and susceptibility to noisy data limit the SVM's performance,

particularly with large datasets. Innovative approaches like the extended kalman filter (EKF) coupled with SVM [9] and adaptive neuro-fuzzy inference system have shown promise in fault classification. Moreover, advanced techniques differential current signal is broken down into distinct energy levels using a novel technique called multi-resolution analysis (MRA), which is then fed to a bidirectional gated recurrent unit (BIGRU) [10]. Furthermore, methodologies such as the dq0 transformation [11] and EKF [12] have been proposed for discriminating between IC and ID. However, these approaches may have limitations, such as neglecting external faults and being tailored for single-phase transformers. In recent studies, the integration of fuzzy logic and entropy measures [13], as seen in multi-scale multivariate fuzzy entropy (MMFE), has shown superiority over traditional approaches like wavelet transform (WT) in distinguishing transformer fault currents from IC. Additionally, innovative fault location approaches utilizing back-propagation neural networks and discrete WT (DWT) [14], as well as ensemble machine learning techniques coupled with internet of things (IoT) systems, have been proposed for transformer defect diagnosis. These advancements highlight the ongoing efforts to enhance transformer protection methodologies, incorporating diverse techniques ranging from harmonic analysis to advanced machine learning algorithms and IoT integration [15], to ensure reliable and efficient operation in power systems. In recent studies, novel methodologies have emerged aiming to enhance transformer protection systems. A probabilistic neural network (PNN) was employed in [16], utilizing discrete wavelet details of the differential current signal to discriminate between IC and ID scenarios in single-phase transformers. However, notable drawbacks of this approach include the absence of explicit mention regarding execution accuracy and decision-making time, as well as its design being tailored solely for single-phase transformers, neglecting the broader application to three-phase systems. Similarly, in a new approach was proposed to discriminate between external and internal defects in PTs by leveraging WT and PNN [17]. Notably, this study failed to consider the effects of IC and current transformer (CT) saturation. Hybrid approaches have also been explored, as demonstrated in [18], where SVM and WT are combined. In this method, the parameters of the SVM are adjusted using particle swarm optimization (PSO), showcasing improved performance. Additionally, a decision tree-based approach is introduced in [19], which claims enhanced accuracy in distinguishing between internal faults and inrush scenarios. Comparative analysis against SVM indicates superior speed and accuracy, with claimed advantages of a decision time of 0.02 s and an accuracy rate of 97.77%. A novel approach is proposed in [20] to fault diagnostics that utilizes WT and deep learning. The suggested method uses WT to extract differential current characteristics and combines long short-term memory

(LSTM) with one-dimensional convolutional neural network (1DCNN) to categorize IDs from other transient phenomena. In [21], a simple yet effective technique for enhancing the functionality of a differential relay is discussed. The proposed method employs the most advanced technique for differential relays available in the current generation, following the classification of the relay's input signal using a classification algorithm. Through DWT, the signal undergoes decomposition into low and high-frequency components via the application of high-pass and low-pass filters [22], followed by a downsampling process to generate approximation and detail coefficients for specific frequency bands. In [23], a time–frequency analysis-based approach is introduced, employing PSCAD/EMTDC for modeling PT operation and IDs. The method utilizes a hyperbolic S-transformer to process simulation data and extract a determination index. Authors in [24] introduce a technique for diagnosing IDs occurring during ICs in PTs. This method utilizes data windows and stacked denoising autoencoders, eliminating the need for a threshold to differentiate between IDs and ICs. Authors in [25] present a hardware-in-the-loop (HIL) test for a transformer differential protection system, verifying the accuracy of the relay's settings configuration. The testing phase includes step-by-step construction to validate the principle of differential computations. In [26] suggest a deep learning protection system based on a CNN, focusing on discriminative features. This network aims to extract distinctive characteristics of the unsaturated part of the differential current, enabling reliable differentiation between IDs and ICs. Authors in [27] propose using the decaying DC component of negative sequence differential current to identify transformer inrush current. Three criteria are considered for setting a threshold to recognize IDs, validated through simulation results for various IDs and inrush situations. In [28], the wavelet packet transform (WPT) and S-transform are employed to identify currents in the PT. Initially, the minimal description length (MDL) requirements within the WPT method guide the selection of the optimal mother wavelet and resolution level for feature extraction from the WPT tree.

Despite the promising outcomes of these approaches, they still exhibit limitations. While they offer advancements, they fall short of achieving complete precision and efficiency. Nonetheless, they represent significant strides towards the development of more robust relay systems for transformer protection.

1.3 Shortcomings

The literature review on recent advancements in transformer differential protection methods provides a comprehensive overview of various techniques and methodologies. However, several shortcomings can be reported as follow:

1. *Limited coverage* The review primarily focuses on summarizing existing approaches rather than critically evaluating their effectiveness or comparing them comprehensively. This limits the depth of analysis and understanding of the state-of-the-art in transformer protection.
2. *Lack of evaluation metrics* While the review mentions the promise and advantages of different methods, there's a lack of specific evaluation metrics or quantitative assessments of their performance. For instance, the effectiveness of each method in terms of accuracy, speed, computational efficiency, or robustness is not thoroughly discussed.
3. *Lack of comprehensiveness* Although some drawbacks are briefly mentioned for certain approaches, such as computational complexity or neglect of external faults, a deeper analysis of the limitations and challenges faced by each method is lacking. Understanding these limitations is crucial for identifying areas of improvement and future research directions.
4. *Insufficient comparison* While some comparative analysis is provided between different methods, such as SVM and decision tree-based approaches, it's not comprehensive enough to draw definitive conclusions about which methods are superior under different scenarios or conditions.

In addressing these shortcomings, future literature reviews could benefit from incorporating a more critical analysis, quantitative evaluation, comparative assessment, and a forward-looking perspective to provide a comprehensive understanding of the field's advancements and limitations.

Table 1 outlines the limitations identified in prior research and studies concerning various parameters, including resistance to noisy data, CT saturation, real-time processing and comprehensiveness in detecting different types of faults.

1.4 Contributions

This study presents a novel method that merges DWT and LSTM networks to improve real-time decision-making processes. By integrating these techniques, the system aims to enhance the precision and dependability of fault detection systems by accurately distinguishing between IDs and ICs

in PTs. LSTM, renowned for its efficacy in nonlinear prediction, serves as the core technique in this methodology. The approach involves decomposing the current signal using DWT to extract various aspects of the wave current, followed by the computation of the energy of three details (ETD), which serves as input data for the LSTM algorithm. Additionally, a supplementary method employing an enhanced derivative approach is introduced to ensure the differential relay's functionality remains intact under CT saturation.

The originality and contribution of this research can be categorized as follows:

1. **Novelty of WT-LSTM integration:** This study pioneers the application of WT-LSTM for discriminating between IC and ID in three-phase power transformers, marking a significant advancement in transformer fault diagnosis techniques.
2. **Utilization of spectral energy:** By leveraging the spectral energy of detail coefficients instead of raw detail coefficients as input data for LSTM, the proposed model effectively minimizes computational burden and facilitates real-time fault detection within each moving window.
3. **Streamlined diagnosis process:** The model streamlines the diagnosis process by integrating traditional methods of signal noise reduction, feature extraction, feature selection, and feature categorization, enhancing the efficiency and accuracy of fault detection.
4. **Enhanced robustness:** Combining WT and LSTM mitigates the limitations associated with conventional approaches, bolstering the model's robustness and ensuring more reliable fault diagnosis outcomes.
5. **Improved performance:** The enhanced algorithm demonstrates superior performance in accurately detecting and diagnosing IC from IDs. It requires fewer training parameters and diagnosis time while exhibiting potent generalization abilities.
6. **Versatile application:** The proposed technique is applicable to fault diagnostics across various capacities and voltage ranges of power transformers, including scenarios with high ground resistances, showcasing its versatility and adaptability.
7. **Comprehensive data collection:** For data training and testing, the model encompasses all transformer cases, including external faults, IC, and ID, spanning a phase

Table 1 Contrasting the characteristics of the suggested approach with those of recently employed methods

	[29]	[6]	[30][31]	[10, 32]	[12, 33]	[34]	[35]	[36]	[37]	This paper
Resistance to noisy data	×	×	×	✓	×	×	✓	✓	×	✓
Resistance to CT saturation	×	✓	✓	✓	×	✓	×	×	×	✓
Real-time processing	×	×	×	×	✓	✓	×	×	×	✓
Detecting different types of faults	✓	✓	✓	✓	✓	×	✓	×	✓	✓

angle range of 0° to 360° , ensuring comprehensive coverage and robustness in real-world applications.

1.5 Organization

The paper comprises five sections, structured as follows: The second section discusses the methodology of study within the proposed model. Details of the suggested technique are presented in the third section. Simulation testing of the model and the subsequent results are outlined in the fourth section. also this section delve into the energy calculation of wavelet details and the processing of variable conditions using LSTM, respectively and evaluation of the performance algorithm and the corresponding results are covered in this section. Finally, the fifth section presents the conclusions drawn from the research.

2 Methodology of Study

As previously discussed, the real-time detection of inner faults in PTs is of paramount importance for ensuring the reliability and safety of power systems. Detecting these faults early enables swift intervention, effectively averting potential damage to the transformer and the broader power infrastructure. Additionally, early fault detection serves to preempt costly repairs and replacements, while also mitigating the economic losses associated with downtime and service

interruptions. the proposed methodology, illustrated in Fig. 1, integrates an enhanced wavelet transform with LSTM neural networks to provide a robust approach for fault detection under these challenging circumstances. The procedural steps of our proposed model, delineated below according to the schematic representation in Fig. 1, are as follows:

a. Data acquisition and preprocessing:

- During transformer energization, sensor data such as voltage and current is meticulously gathered.
- Preprocessing entails filtering and normalization procedures to optimize the data for subsequent analysis. Pre-processing entails filtering and normalization procedures to optimize the data for subsequent analysis.

b. Enhanced wavelet transform:

- Building upon conventional techniques, an improved WT is applied to the preprocessed data.
- This transformation dissects the signal into distinct frequency components, effectively capturing transient and steady-state features.
- Notably, the wavelet-transformed data accentuates variations indicative of fault conditions amidst ICs.

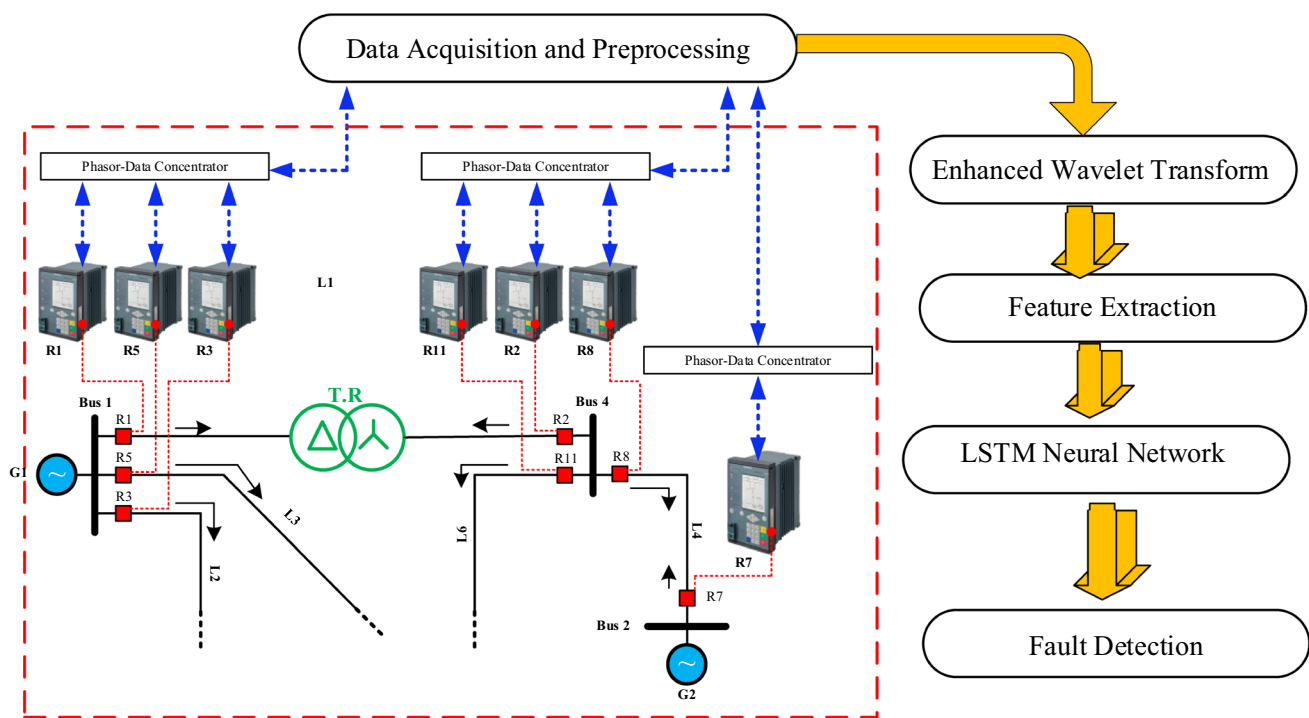


Fig. 1 Conceptual model of proposed real-time inner fault detection

c. Feature extraction:

- From the wavelet-transformed data, pertinent features are meticulously extracted.
- These features serve to characterize the transient behavior of the signal during instances of ICs.
- Statistical measures and energy distribution across frequency bands represent some of the features that may be extracted.

d. LSTM neural network:

- Leveraging the extracted features, an LSTM neural network undergoes training.
- LSTM networks are particularly helpful for analyzing time-series data, such as ICs, because of their well-known ability to capture temporal relationships in sequential data. Because of the way they are made, they can hold onto information for a long time, which is important when it comes to seeing trends and abnormalities in data that changes over time.
- The network's learning process encompasses discerning patterns indicative of both normal operation and fault conditions amid ICs.

e. Training and evaluation:

- The LSTM network undergoes rigorous training utilizing labeled data, where the labels denote fault conditions observed during ICs.
- Training iteratively refines the network's parameters to minimize the disparity between predicted and actual fault conditions.
- Evaluation metrics such as accuracy, precision, and recall are employed to gauge the fault detection system's performance.

f. Fault detection:

- Upon completion of training, the LSTM network is deployed for real-time fault detection amidst ICs.
- Incoming sensor data is fed into the network, enabling it to promptly classify transformer's condition as normal or indicative of a fault.
- Swift fault detection during inrush currents facilitates timely intervention, thereby mitigating the risk of poten-

tial damage to the transformer and the wider power infrastructure.

2.1 Theoretical and Analysis of Inrush Current and Internal Fault

Surge currents happen when a transformer is triggered, usually during startup or when it is re-energized. These currents have the ability to momentarily conceal systemic flaws. An extensive examination of the surge current and its source will be conducted in order to more clearly distinguish between surge currents and internal problems. In this analysis, the effects of the transformer and circuit resistance will not be taken into account when calculating the supply voltage using Eq. (1) [38].

$$u = U_m \sin(\omega t + \alpha) \quad (1)$$

When a no current transformer in the primary coil is connected to the mains, it creates a link between the core's magnetic field and voltage input. The behavior governs how the magnetic material surrounding the conductor reacts when electrical energy is sent via the equipment:

$$N_w \frac{d\varphi}{dt} = U_m \sin(\omega t + \alpha) \quad (2)$$

In this equation N_w represents the number of winding turns on the no-load side of the transformer, φ denotes the magnetic flux within the iron core, U_m signifies the peak value of the supply voltage, and α represents the switching angle.

If N is set to 1, the indeterminate Eq. (2) can solve as following:

$$\varphi(t) = -\frac{U_m}{\omega} \cos(\omega t + \alpha) + C \quad (3)$$

where When $t=0$, the remnant flux φ_r that is present under initial conditions determines C , the integration constant.

$$C = \frac{U_m}{\omega} \cos(\alpha) + \varphi_r \quad (4)$$

When a no-load transformer is input, the magnetic flux is typically formulated by Eq. (5).

$$\varphi(t) = -\varphi_m \cos(\omega t + \alpha) + \varphi_m \cos(\alpha) + \varphi_r \quad (5)$$

The exciting current $i_e = 0$ stays zero when the magnetic flux is below φ_s because the iron core does not reach saturation. However, the inductance L is assumed to remain constant once the magnetic flux surpasses φ_s and the iron core becomes saturated. The constant value of L can then be used to calculate the inrush current under these circumstances.

$$\begin{aligned}
i_e &= \frac{\varphi(t) - \varphi_s}{L} = \frac{U_m}{\omega L} [\cos \alpha - \cos(\omega t + \alpha)] + \frac{\varphi_r - \varphi_s}{L} \\
&= \frac{U_m}{\omega L} \left[\cos \alpha - \cos(\omega t + \alpha) + \frac{\varphi_r - \varphi_s}{\varphi_m} \right] \\
&= \frac{U_m}{x} \left[\cos \alpha - \cos(\omega t + \alpha) + \frac{B_s - B_r}{B_m} \right] \geq 0
\end{aligned} \quad (6)$$

The saturated flux φ_s , the remnant flux φ_r , and the switching angle α all have an impact on the inrush current. As can be seen in Fig. 2, we will be using a data window (DW) diagram for both IDs and ICs in the forthcoming discussions. This figure displays a waveform inside the DW that draws attention to particular features at every sampling point.

A LSTM network needs these differentiating characteristics in order to distinguish between IDs and ICs. A portion of the instantaneous differential currents in a DW during a single cycle are shown in Fig. 2, which also displays the conditions of internal faults and inrush currents. Until the circuit breaker is tripped, all values in the DW are initially at zero. This causes a sudden change in magnetic flux and a notable increase in differential current. When inrush currents and internal faults occur simultaneously, it can be difficult to differentiate between them using conventional methods because an internal fault displays a similarly abrupt initial change in differential current. Before point A in the DW in the example shown in Fig. 2, all values are at a baseline of zero, indicating a normal condition (referred to as state 0). The last index in the DW changes to a positive value at point A, denoting the start of a temporary phase (called state 1). Identification and distinction of these disturbances become difficult when an IC and an ID occur at this point, frequently leading to delayed responses. As we move on to point B, we see that IDs exhibit different behaviors. At an angle of 0° or 90° , the value either returns to zero or shifts to the negative. When the algorithm recognizes these particular patterns, it immediately moves to state 3, signaling the occurrence of

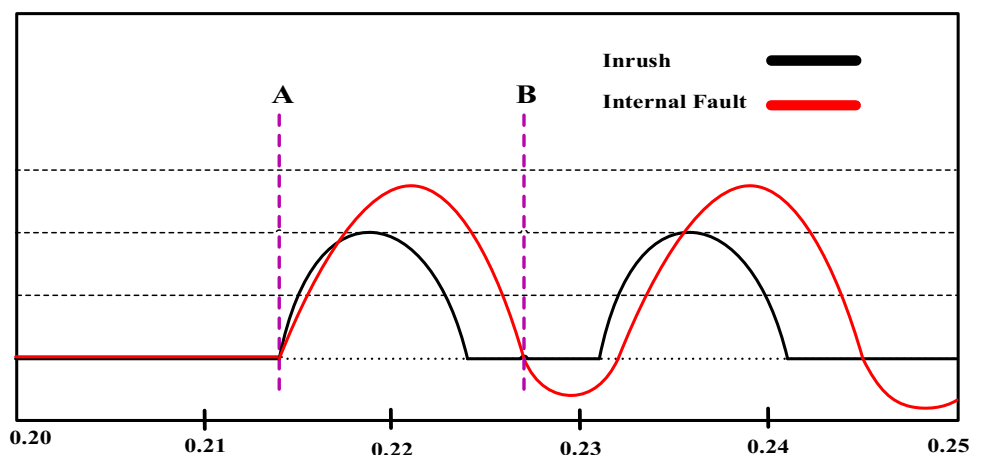
an ID. When these identifiable patterns are absent, the state of affairs is categorized as an inrush current (state 2). There are 58 samples in the time interval between points A and B, which is marginally less than a full cycle of 64 samples. The precise distinction between IDs and ICs depends on this sampling delay, particularly in situations where the fault has a 0° inception angle.

2.2 Modified Discrete Wavelet Transform (MDWT)

When it comes to processing non-stationary signals, WT provides a stable method for time–frequency domain analysis that has clear advantages over fast Fourier transform (FFT). By splitting the signal into approximate and detail components, the DWT reveals a variety of signal characteristics and is shown to be a highly efficient method of extracting signal information. The application of high-pass and low-pass filters, which divide the signal into high and low frequency bands, respectively, facilitates this segmentation process. These filtered outputs are then subjected to a down-sampling process to produce approximation and detail coefficients, each of which corresponds to a distinct frequency range. Recursive processing is applied to the low-pass filter's output, as shown in Fig. 3, up until a predetermined point of decomposition. With this method, all of the frequency components of a transient signal that has been simplified by the transformation process can be precisely manipulated. DWT breaks down the signal $x(t)$ into low-frequency (approximation, A) and high-frequency (detail, D) components in the context of fault current analysis, successfully encapsulating the signal's basic dynamics.

$$x(t) = \sum A_0 \Phi_{j,k}(t) \quad (7)$$

Fig. 2 Demonstration of a data window capturing differential current during inrush and internal fault scenarios



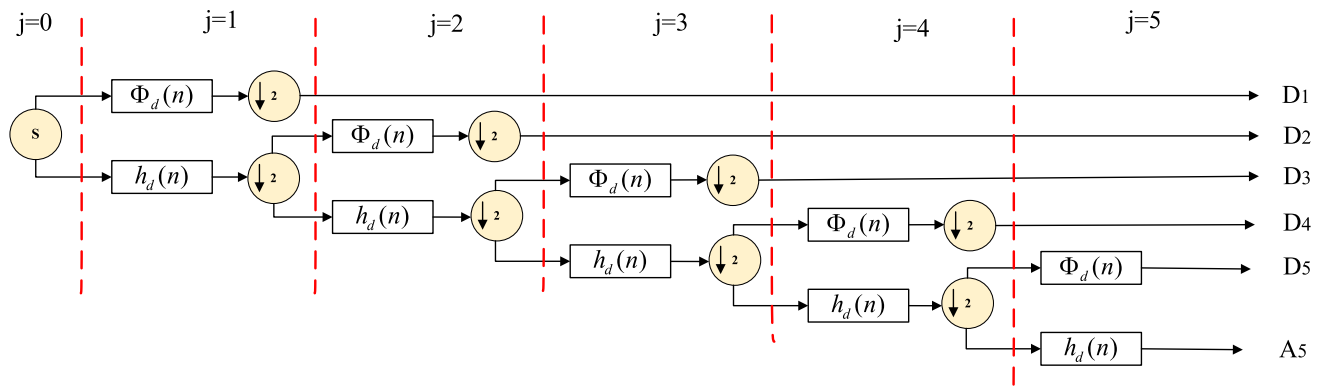


Fig. 3 Multi resolution for approximation and details

$$x(t) = \sum_k A_1 \Phi_{j-1,k}(t) + \sum_k D_1 \Phi_{j-1,k}(t) \quad (8)$$

$$x(t) = A_1(t) + D_1(t) \quad (9)$$

$\Phi_{j,k}(t)$, which is important in wavelet transformation analysis, represents either the parent wavelet or the base function in the given formulas. D_1 denotes the detailed coefficients that cover the higher frequency features of the signal, while A_0 and A_1 represent the rough coefficients that represent the lower frequency elements of the signal. The process of breaking down the low-frequency component, also known as the approximation coefficients, is carried out through step N . Through this dissection, the essential signal details can be separated from any underlying interference. Equations (10–12) elaborate on the particular steps and mathematical computations involved in this disintegration. These equations methodically demonstrate how each approximation step improves the signal analysis by separating and elucidating the critical information from the interference elements.

$$x(t) = A_2(t) + D_2(t) + D_1(t) \quad (10)$$

$$x(t) = A_3(t) + D_3(t) + D_2(t) + D_1(t) \quad (11)$$

$$x(t) = A_N(t) + D_N(t) + D_{N-1}(t) + \dots + D_1(t) \quad (12)$$

The parameter N indicates the depth of the decomposition during the signal breaking process, meaning how closely the signal is examined. With each new level, the analysis of the frequency components is extended, revealing ever-more-detailed characteristics of the signal. The upper bound on the decomposition levels, represented by the letter L , is determined based on certain standards in order to maintain the integrity of the analysis and avoid going beyond what

the resolution support of the data will allow. Equation (13), which establishes limits on the highest level L that can be used for decomposition in order to guarantee a reliable and effective analysis free from overfitting or redundancies in the signal processing process, formally expresses this constraint. Following these guidelines ensures a complete and precise decomposition process, which facilitates the best possible extraction of insightful information from the signal.

$$N = 2^L \quad (13)$$

The signal decomposition procedure is mathematically captured in Eq. (14) which shows how the signal is divided into discrete frequency bands at each level. When high-pass and low-pass filters are used together, the signal is efficiently broken down into its individual components, which allows for a thorough examination of its frequency spectrum. The approximation coefficients from the low-pass filters capture the signal's general low-frequency characteristics, while the detail coefficients from the high-pass filters capture the complex high-frequency features. This segmentation makes it easier to isolate and analyze the different frequency components that make up the fault current signal, which enables a more in-depth investigation of its characteristics and behaviors over a wider frequency range. By using this methodical breakdown technique, important information about the dynamics, patterns, and anomalies of the fault current can be obtained, which advances our comprehension of the underlying phenomena and improves our capacity for fault detection and diagnosis.

$$B = \frac{F}{2^{L+1}} \quad (14)$$

Here, B stands for the bandwidth (measured in Hertz) for each level, and F stands for the sampling frequency (measured in Hertz as well).

It is important to evaluate the variety of mother wavelets available to you when choosing from the enhanced WT, such as Haar, Morlet, Daubechies (db), Coiflet, and Symmlet. The orthogonality of Haar and Morlet wavelets is recognized, whereas the non-orthogonality of Symmlet and Daubechies is acknowledged. Due to its efficiency and versatility, the Daubechies wavelet is one of the most widely used in a variety of power system scenarios. The Daubechies wavelets, sometimes referred to as db wavelets, are very good at identifying signals that exhibit oscillatory behavior, short duration, low amplitude, and fast decay—all of which are frequently present in power system disruptions. Wavelets from the db family with filter coefficients of 4, 6, 8, and 10 are useful in a variety of disturbance detection situations. Nevertheless, the particular needs of the application will determine which wavelet is best. For example, in power system analysis, longer-lasting and slower-moving transient disturbances are better detected with db8 and db10 wavelets, while short-lived transient disturbances are best identified with db4 and db6 wavelets. When there are fast disturbances, such as faults, the waveforms often show quick changes, abrupt transitions, and sudden variations within brief intervals. The db4 wavelet has been widely researched in fault detection studies and is very effective at accurately capturing these quick disturbances. Researchers have extracted features like standard deviation (SD) and energy values (EVs) from the db4 mother wavelet in the current study, which focuses on identifying internal faults. A sampling rate of 20 kHz and a time span of 5000 have been chosen for this analysis so that DWT can analyze the current signal. In order to efficiently record the swift oscillations in fault current signals, a sampling frequency of 20 kHz is utilized. However, there are difficulties when identifying internal power transformer faults when using the wavelet-transformed signal as the direct input for training the LSTM network. The ideal number of sampling points, represented by the letter “n,” must be carefully chosen in order to achieve effective feature extraction. This procedure is essential for developing a computational intelligence-based detector that can detect internal flaws. Researchers usually choose to build the dataset needed to train the fault detection system over a time interval of one to three cycles of the current signal’s power frequency. The present study makes use of features that are extracted using DWT analysis in order to identify IDs in PTs. But problems occur when applying SD features that are computed over a quarter or a full cycle of the power frequency when there is a fault, which could result in misidentification. In order to improve fault detection accuracy, features like SD and EVs are calculated over a longer time interval that includes two power cycles (800 samples) during the pre-fault, fault, and post-fault phases. This more comprehensive method enables a detailed analysis of fault dynamics. In order to accurately classify the fault type, the DWT analysis is performed

iteratively over a duration of 0.04 s (equate to 800 samples). As shown in Table 2, the frequency range that is captured at each decomposition level varies and is determined by Eq. (6). Transient signals decrease as the decomposition levels rise, and by level 5, they are completely eliminated. It has been shown that this methodological approach works well for increasing the fault detection process’ resilience.

2.3 Energy Value Calculation

The EV is calculated by adding together the squares of the detail coefficients (D1 to D5) and the approximation (A5) from the fault current signal decomposition. The method for determining EV for each phase is as follows:

$$EV_{abc} = \left(\sum_{i=1}^k |D_i^a|^2 + |A_5^a|^2 \right) + \left(\sum_{i=1}^k |D_i^b|^2 + |A_5^b|^2 \right) + \left(\sum_{i=1}^k |D_i^c|^2 + |A_5^c|^2 \right) \quad (15)$$

In this context, the number $k=5$ corresponds to the levels, where the detail coefficient levels D1, D2, D3, D4, and D5 are identified, and the approximation of the fifth level of the signal is denoted as A5. Analogous to the SD value, lower values of EV are typically observed during fault transients in contrast to regular system functioning. As a result, EV functions as a characteristic utilized for distinguishing the nature of disturbance present in the system.

2.4 Long Short-Term-Memory Network

As discussed earlier, the primary aim of this paper is to create a computational model employing an LSTM neural network to precisely detect inner faults in power transformers. This model will rely on features extracted from fault current signals transformed using wavelets to distinguish between normal operation and the existence of inner faults. The significance of employing LSTM networks for identifying inner faults in power transformers is notable for various reasons:

Table 2 Frequency range of detailed coefficients in improved wavelet transform

Degree of granularity coefficient	Minimum bandwidth in kilohertz (kHz)	Maximum bandwidth in kilohertz (kHz)
D ₁	5	2.5
D ₂	2.5	1.25
D ₃	1.25	0.625
D ₄	0.625	0.3125
D ₅	0.3125	0.15625

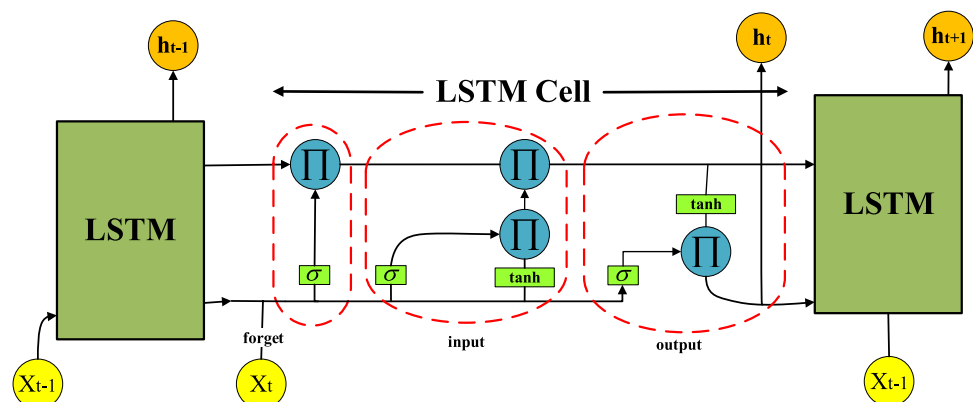
1. *Temporal dependency* LSTM networks are adept at capturing temporal dependencies in sequential data. In the context of power transformers, fault signals often exhibit complex temporal patterns that evolve over time. LSTM networks can effectively model these temporal dynamics, enabling accurate identification of inner faults.
2. *Handling time-series data* Power system data, including fault current signals, is inherently time-series data. LSTM networks are specifically designed to handle such data, making them well-suited for analyzing fault signals collected from PTs.
3. *Long-term memory* Traditional neural networks may struggle to retain information over long sequences, which is crucial for identifying faults occurring in power transformers. LSTM networks, with their ability to maintain long-term memory through gated mechanisms, are better equipped to handle such scenarios.
4. *Feature extraction* LSTM networks are highly skilled at autonomously extracting essential characteristics from unprocessed data, a valuable ability particularly beneficial in scenarios where distinguishing significant attributes from intricate signals is difficult, like in fault detection. This automatic feature extraction boosts the system's effectiveness in recognizing intricate patterns and irregularities within the data, ultimately streamlining the evaluation process and enhancing overall performance.
5. *Adaptability to various fault conditions* Inner faults in PTs can exhibit diverse characteristics, including different fault types, magnitudes, and durations. LSTM networks can adapt to these variations and learn to distinguish between normal operation and various fault conditions.
6. *Real-time detection* Prompt detection of inner faults is crucial for preventing catastrophic failures and minimizing downtime in power systems. LSTM networks can be deployed for real-time fault detection, providing timely alerts and enabling proactive maintenance strategies.
7. *Robustness to noisy data* Power system data often contains noise and disturbances, which can hinder fault detection accuracy. LSTM networks are inherently robust to noisy inputs and can effectively filter out irrelevant information, improving the reliability of fault identification.
8. *Continual learning* Power system conditions may evolve over time due to factors such as load fluctuations and environmental changes. LSTM networks support continual learning, allowing them to adapt to changing operating conditions and maintain high detection accuracy over extended periods.

Within this framework, the LSTM recurrent neural network structure enhances its capacity to handle large input sequences by integrating memory cells and gate mechanisms. These components address the common issues associated with training conventional RNNs, such as struggles in capturing extended dependencies. The memory cells enable the network to preserve information for extended durations, while the gates regulate the information flow in and out of the cell, optimizing the learning process control. The memory cell establishes a continuous pathway across the network, facilitating the connection of multiple LSTM cells without encountering the “vanishing gradient issue” prevalent in conventional RNN setups. Additionally, to update the memory state within this cell, three gates are employed:

- The forget gate regulates the retention of past information by the LSTM.
- The input gate governs the incorporation of new data into the LSTM's memory.
- The output gate controls the memory utilization in generating the output at the current time step.

Figure 4 illustrates the internal architecture of the LSTM, illustrating the operation of these components [39]. The following expression can be used to express the computation of LSTM states at a given time step t :

Fig. 4 Internal organization of LSTM



$$f_t = \sigma(W_f \times X_t + U_f \times h(t-1)) + b_f \quad (16)$$

$$i_t = \sigma(W_i \times X_t + U_i \times h(t-1)) + b_i \quad (17)$$

$$O_t = \sigma(W_o \times X_t + U_o \times h(t-1)) + b_o \quad (18)$$

$$c_t = f_t \otimes C(t-1) + i_t \otimes \tanh(W_c \times x_t + U_c \times h(t-1) + b_c) \quad (19)$$

$$h_t = O_t \otimes \tanh(c_t) \quad (20)$$

where X_t =the input of vectors; f_t =The activation vector of the forget gate at time t ; i_t =receive the activation vector for the input gate at time t ; O_t =activation vector for the output gate at time t ; c_t =The vector that activates the memory cell at a specific time point; h_t : The vector that triggers the hidden state memory at a given moment t ; W_f, W_i, W_o : the input filters of the pertinent gates; U_f, U_i, U_o : the corresponding gates' recurrent kernels; b_f, b_i, b_o : the biases; σ :the sigmoid function of logistics; $\otimes$:the product of matrix; $\tanh$ = the activation function of hyperbolic tangent.

The training of the model involves utilizing an appropriate loss function, like binary cross-entropy, to reduce the discrepancy between the predicted output $\hat{Y}$ and the true labels Y . The network's parameters θ are adjusted through techniques like backpropagation through time (BPTT) or optimization algorithms such as Adam and RMSProp.

Where Y : Output labels indicating the fault condition (0 for normal operation, 1 for inner fault); θ : Parameters (weights and biases) of the LSTM network.

The LSTM network being utilized comprises the following layers, as illustrated in Fig. 5:

- **Sequence input:** This layer is designed to handle sequential data, implying that the data presented to the neural network possesses an inherent order or temporal aspect, which is particularly pertinent for tasks such as time series forecasting.
- **LSTM layer:** This layer constitutes a pivotal component enabling the network to process and comprehend time series or any sequential data. LSTMs are tailored to retain and discern patterns across extended sequences and exhibit reduced susceptibility to the vanishing gradient issue that impacts RNNs.
- **Fully connected (FC) layer:** The fully connected layer establishes connections between every neuron (or unit) in the preceding layer with every other neuron in that layer. Serving as a bridge between the regression output and the LSTM layer, this layer ensures the transformation of the high-dimensional output from the LSTM into a suitable regression format.
- **Regression output layer:** This particular layer is the last element in a neural network designed for regression. Unlike scenarios involving classification, where the goal is to allocate input data into predefined categories, regression tasks focus on forecasting continuous values instead.

3 Proposed Technique Implementation

To validate the precision and efficacy of the proposed methodology, a comprehensive power system was constructed and subjected to testing. Comprising elements such as a three-phase power supply, a 3-phase power transformer rated at 33/11 kV with a capacity of 31.5 MVA

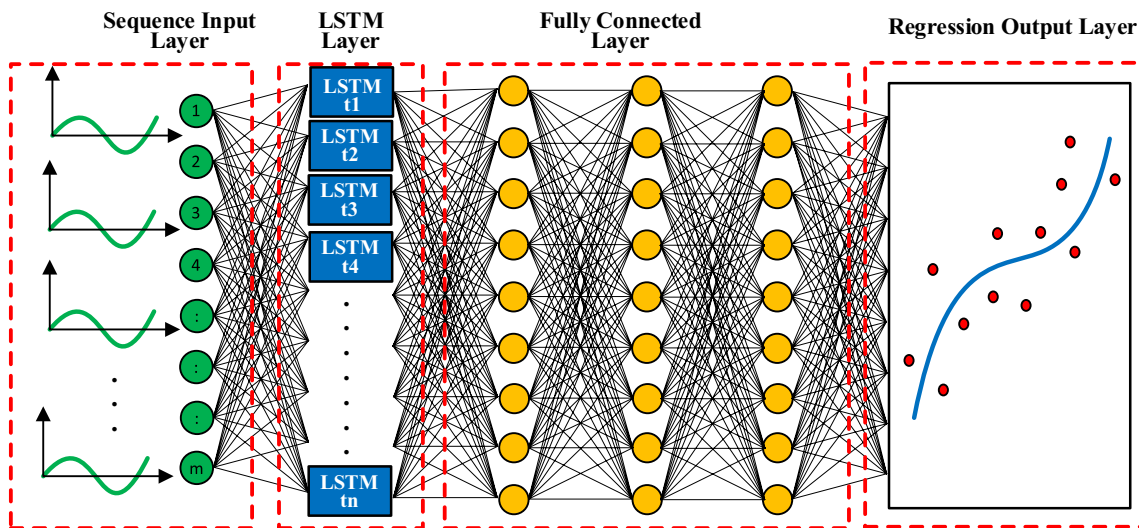


Fig. 5 LSTM neural network architecture

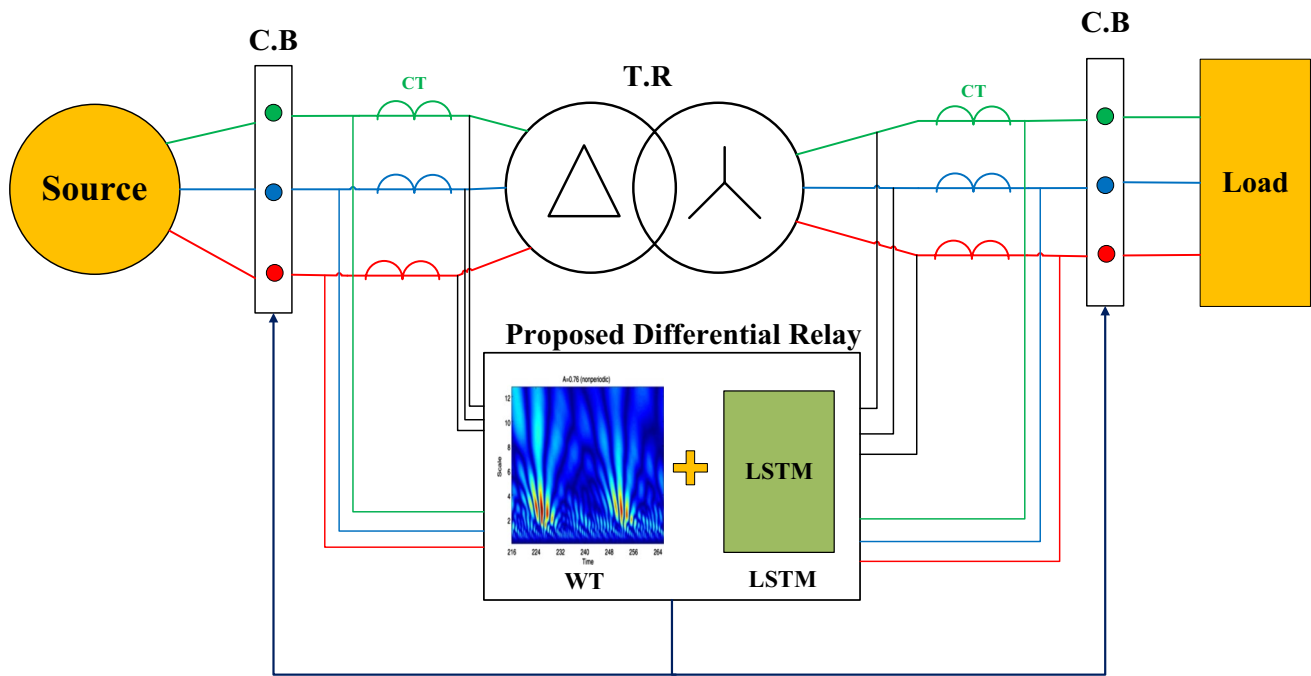


Fig. 6 Configuration of the proposed differential relay in power system

Table 3 Data of power system proposed model

Connection of Transformer	Dyn11
Rated power	31.5 MVA
Voltage ratio	33/11 kV
Rated frequency	50 Hz
Percentage impedance	10%
CT ratio primary side	600/5
CT ratio secondary side	2000/5

Table 4 Specifications of LSTM algorithm

Specification	LSTM
Function of training	sgdm
Total number of layers	three
Number of LSTM layers	one
Initial learn rate	0.01
Maximum epochs	200
Learn rate drop factor	0.1

current transformers positioned on both sides of the PT, along with three-phase circuit breakers as depicted in Fig. 6. Detailed datasets pertaining to the transformer characteristics are delineated in Table 3, while Table 4 provides an exhaustive overview of the functions and specifications of the LSTM employed in the analysis.

The simulation of the power system is conducted using MATLAB/SIMULINK software, with the procedural workflow outlined in the following algorithm. The steps involved in creating the model are as follows:

Step 1 Differential current measurements are taken for the three phases, followed by the application of wavelet transform for feature extraction.

Step 2 Spectral energy is calculated for the three details obtained from the wavelet transform.

Step 3 LSTM is applied to the dataset for analysis.

Step 4 The dataset is saved, and training, validation, and testing sets are generated.

Step 5 The trained model is verified using the test dataset, and evaluations are conducted to distinguish between ICs and IDs.

operating at 50 Hz frequency, configured in Dyn11 setup, alongside a three-phase load, the system also incorporates

1.	Set $t + \Delta t \rightarrow t$
2.	Input: $I_a(t), I_b(t), I_c(t)$ % Primary side of power tranformer
3.	Wavelet Function ($I_a(t), I_b(t), I_c(t)$): $x_a(t) = A_3^a(t) + D_5^a(t) + D_4^a(t) + \dots + D_1^a(t)$ $x_b(t) = A_3^b(t) + D_5^b(t) + D_4^b(t) + \dots + D_1^b(t)$ $x_c(t) = A_3^c(t) + D_5^c(t) + D_4^c(t) + \dots + D_1^c(t)$
4.	Energy Function ($\{D_1^a, A_5^a, D_1^b, A_5^b, D_1^c, A_5^c\}$): $4.1 \quad EV_{abc} = \left(\sum_{i=1}^k D_i^a ^2 + A_5^a ^2 \right) + \left(\sum_{i=1}^k D_i^b ^2 + A_5^b ^2 \right) + \left(\sum_{i=1}^k D_i^c ^2 + A_5^c ^2 \right)$
5.	LSTM Function (EV_{abc}): $6.1. \quad f_t = \sigma(W_f \times X_t + U_f \times h(t-1)) + b_f$ $6.2. \quad i_t = \sigma(W_i \times X_t + U_i \times h(t-1)) + b_i$ $6.3. \quad O_t = \sigma(W_o \times X_t + U_o \times h(t-1)) + b_o$ $6.4. \quad c_t = f_t \otimes C(t-1) + i_t \otimes \tanh(W_c \times x_t + U_c \times h(t-1) + b_c)$
6.	if $t < t_{\text{Simulation}}$ Classification: -Inrush Current -Internal Fault - External fault - Internal Fault During Inrush current End if
7.	End

3.1 Feature Extraction by Wavelet Transform Analysis

To induce CT saturation conditions and evaluate the efficacy of the improved WT in extracting features from IC, an IC with a magnitude 7–8 times higher than the nominal value is generated on the primary side of the specified transformer. ICs exhibit a rapid increase in magnitude at the onset of space gaps, transitioning either from zero to a significant value or from a specific value back to zero. This abrupt change in current results in smooth ripples along the signal's length as it swiftly shifts between different states. Upon inspecting the shape of the decomposition details depicted in Fig. 7, both the origin and detail coefficients of the IC signal reveal discernible features. These features are indicative of distinct characteristics within the signal, which can be effectively captured and utilized for further analysis and diagnosis.

In the presented scenario, an ID involving three-phase and ground faults (ABC-G) has occurred on the low-voltage side of the PT. Figure 8 illustrates the current signal during this fault occurrence. It is apparent from the figure that due to the capacitances and inductances of transmission lines, the current signal exhibits distortion. Specifically, the origin and detail of the fault current manifest as sharp spikes at the time of the fault in Fig. 8. Upon comparing Fig. 7 with Fig. 8, it becomes evident that there are discernible differences. In contrast to the behavior observed in the absence of a IC, where numerous sharp spikes persist with little attenuation and fail to decay to zero, the sharp spikes of current in the presence of the fault instantaneously decay to zero. Additionally, an area of details appears to be clean with only one spike. This discrepancy between the two figures underscores the distinct characteristics of fault-induced current spikes compared to those occurring under normal conditions.

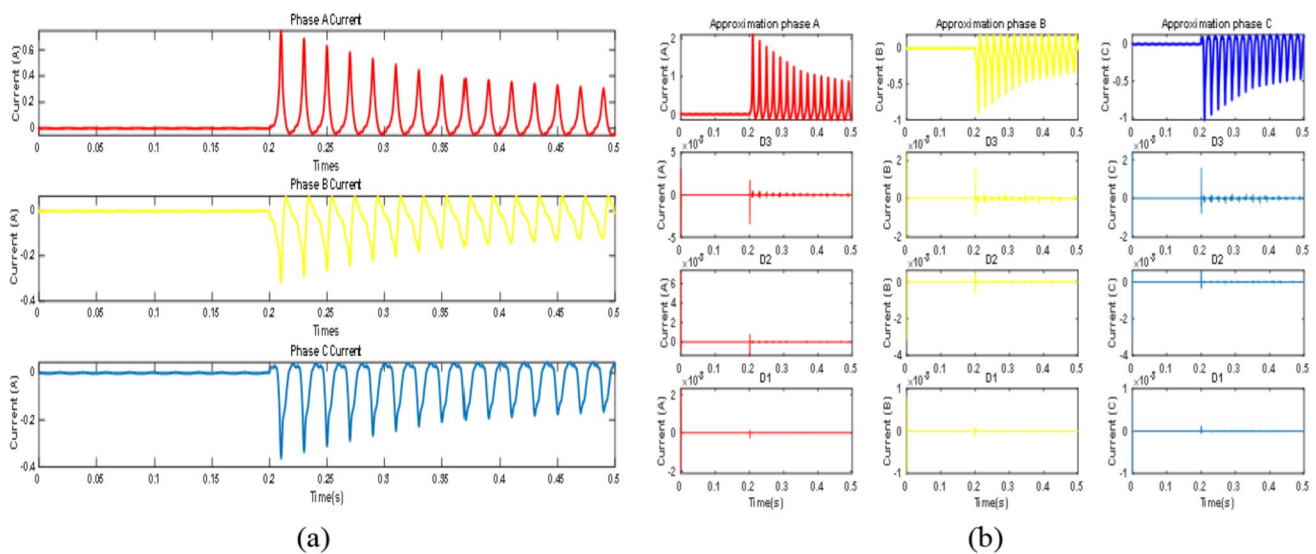


Fig. 7 **a** Origin inrush current **b** Details of inrush current

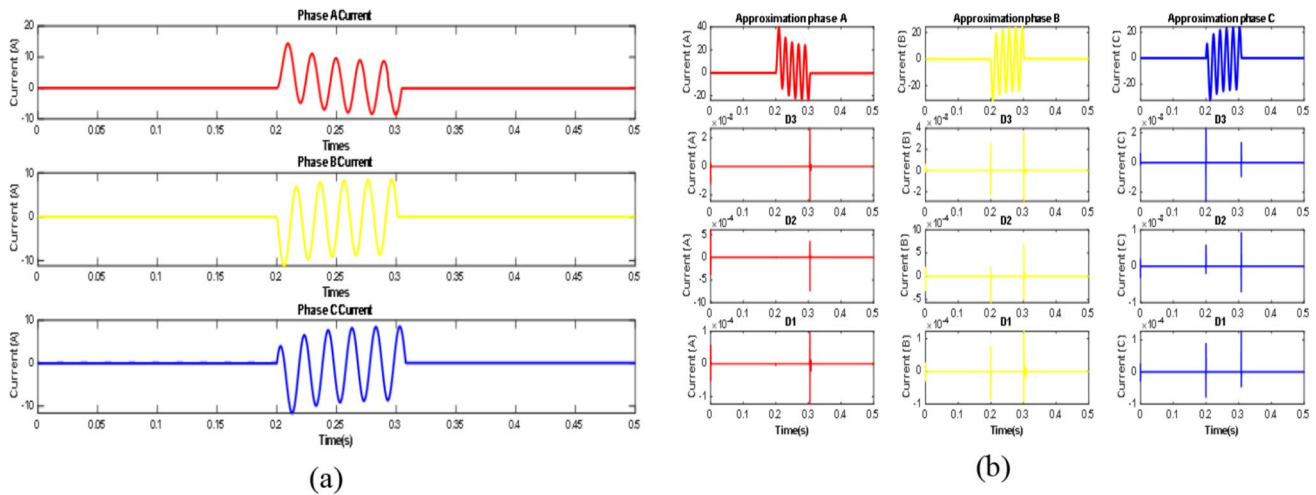


Fig. 8 **a** Origin current **b** Details of internal fault current (ABC-G)

It is important to note that the differential protection settings are configured to address all types of external faults, especially those occurring under CT saturation conditions. In the event of an external fault involving three-phase and ground faults (ABC-G) outside the protected zone of the PT (such as three-phase faults occurring on the load side), wavelet analysis of the current signal was conducted with a ground fault resistance of 33 Ω .

As evident in Fig. 9, both the origin and details of the WT reveal a concentration of sharp spikes during the duration of CT saturation. Notably, these spikes maintain the same shape and level throughout the duration of the faults. These distinct features serve to differentiate external faults from

other scenarios, providing valuable insights for fault detection and protection measures.

3.2 Energy Calculation of Wavelet Details

In this current study, the EVs criterion, as introduced in Eq. (15), is employed to analyze the content of the current signal utilizing the improved WT. To maintain the original properties of the wavelet signal, feature selection is conducted using the spectral energy of detailed coefficients (D1–D5). This approach ensures that the essential characteristics of the wavelet signal are preserved while facilitating effective feature selection for further analysis.

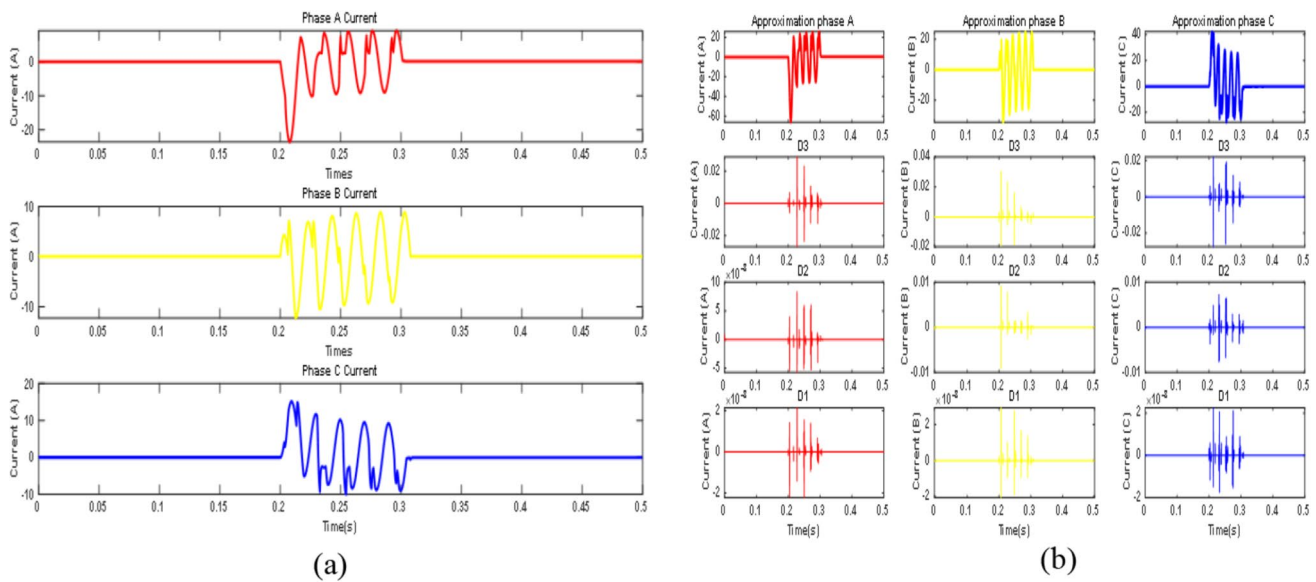


Fig. 9 a Origin current b Details of transformer external fault under CT Saturation (ABC-G)

3.3 Distinguish Variable Conditions Process Using LSTM

Algorithm based on LSTM demonstrates the capability to swiftly and accurately distinguish between IC and IF in PT. LSTM is employed to establish the association between distinguishing features, leveraging its capacity to mitigate the drawbacks of conventional algorithms and achieve higher differentiation accuracy. To utilize LSTM for discrimination, a training dataset is employed to train and evaluate the model. Subsequently, the trained model is applied to identify discrimination within real-world data. In this study, a unique technique is proposed for diagnosing between ICs and IDs, integrating WT and LSTM. The fusion of these two criteria is anticipated to enhance the classification methodology. The implementation of LSTM typically involves several steps:

1. Preprocessing of acquired data: This involves preparing input data for use in the LSTM model, encompassing processes such as feature extraction, normalization, and data cleaning.
2. Designing the LSTM architecture: Determining the number of LSTM layers, the size of LSTM cells, and assessing whether additional layers or components are necessary for the specific application.
3. Training the model: The LSTM model is fed with labeled data, and backpropagation or other optimization techniques are employed to adjust the biases and weights of the LSTM cells. Loss functions, such as cosine, are utilized to minimize errors during training.
4. Analyzing the model's performance: Test data is employed to evaluate the efficacy of the trained LSTM

model in discrimination. This analysis aids in assessing the model's performance in categorization and determining its suitability for the given scenario.

4 Results Analysis and Discussion

As discussed in the preceding section, a PT with specifications (33/11 kV, 31.5 MVA, 50 Hz) configured as Dyn11 is utilized to validate the proposed inner fault detection methodology. The discussion of the DWT-LSTM model's results is founded on testing the model across multiple fault and IC scenarios. The model's performance is evaluated based on training it on the proposed deep learning framework, using a total of 536 cases segregated into four groups: ICs, IDs, external faults, and normal operation. The proposed methodology initially extracts features from the differential current using WT and subsequently computes the spectral energy for three distinct details. As previously mentioned, spectral analyses were conducted for each of the four cases, with each case demonstrating unique characteristics, highlighting the effectiveness of the deep learning model in accurately recognizing various fault conditions.

4.1 Training Procedure of LSTM

In the present analysis, MATLAB 2022a programming software is utilized to conduct experiments using data collected from the model's test, comprising 536 cases representing ICs, IDs, external faults, and normal operation scenarios. The analyses conducted in this study are based on a well-established dataset. During the training phase, the LSTM

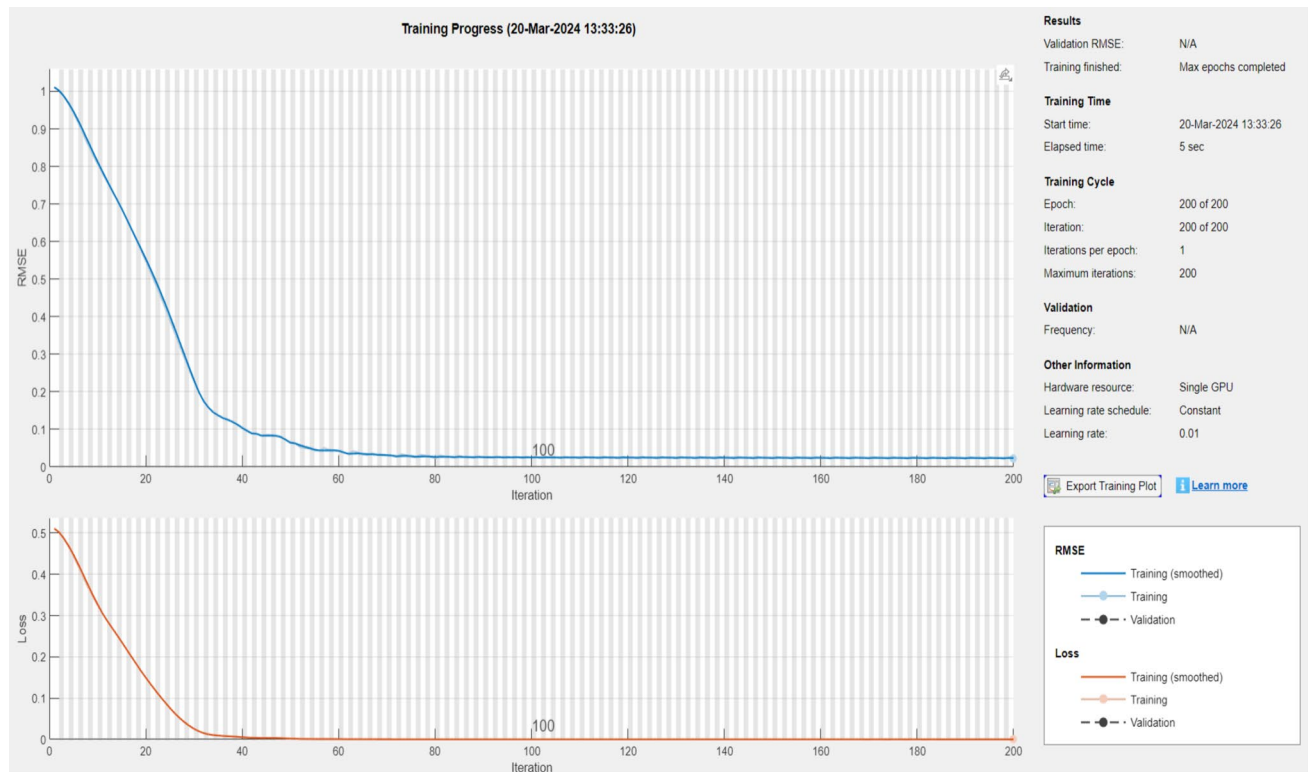


Fig. 10 The validation and training of the LSTM network based on Sgdm optimizer

Table 5 Evaluation of Sgdm optimizer-based on LSTM network's training and validation accuracy at each epoch and iteration level, displaying the associated losses and base learning rate

Epoch	Iteration	Time elapsed (Se)	Mini-batch RMSE	Mini-batch Loss	Base learning rate
1	1	0.02	1.01	0.5	0.010
50	50	0.03	0.06	0.002	0.010
100	100	0.04	0.02	0.003	0.010
150	150	0.05	0.02	0.0024	0.010
200	200	0.05	0.02	0.0027	0.010

network adjusts its weights and biases to minimize prediction errors, thereby gaining the ability to identify patterns and temporal dependencies within sequences of inputs. This is achieved through the distinctive design of LSTM, which incorporates gates and cell state mechanisms. The network is trained to predict outputs, with the raw numbers predicted by the model serving as the output.

The training process comprises 200 epochs, during which the learning rate gradually decreases from its initial value of 0.01 after 200 epochs. The correctness of the network is validated by defining validation data and frequency, with the algorithm regularly assessing the accuracy of validation data throughout the training process. The performance of the

model is evaluated using root-mean-square error (RMSE) of projected images and the percentage of predictions falling within an acceptable error margin. The training and analysis of the proposed LSTM network utilizing the Sgdm optimizer is depicted in Fig. 10. Furthermore, Table 5 illustrates the iteration-wise mini-batch accuracy for both training and validation, alongside the basic learning rate and associated losses. The model achieved a loss accuracy of 2.7×10^{-4} and an RMSE accuracy of 0.02 by the end of the training epoch.

4.2 Classification Results

Utilizing (WT-LSTM) networks presents an innovative approach in sequence classification tasks, particularly when input data comprises both spatial and temporal features. Here's a comprehensive overview of the methodology for fault classification from ICs using WT-LSTM:

Initially, depending on the data's characteristics, segmentation and preprocessing are crucial steps. Once training is completed, the test data is introduced to the model. The outputs of the model, including the first LSTM layer and the final hidden layer, are then showcased after dimension reduction. This process effectively diminishes the overlapping portions of each category, thereby enhancing the recognition rate. Although the distances between samples may appear reasonably close in the final stage, the true

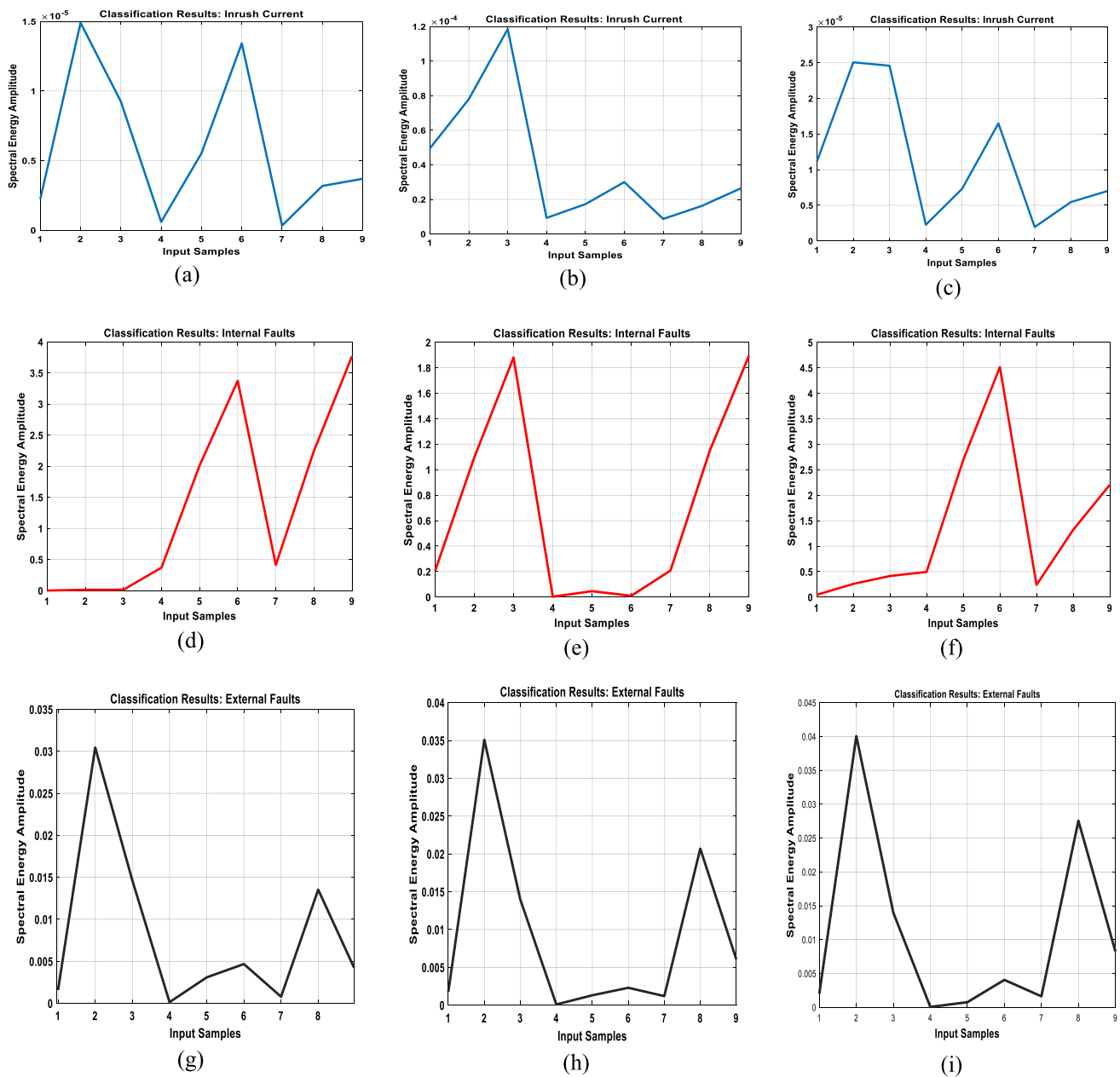


Fig. 11 a–c Inrush current: d–f Different internal faults: g–i Different external faults under CT saturation

classification outcome becomes apparent with the utilization of LSTM. To validate the accuracy and efficacy of the algorithm in discriminating between ICs, IDs, and external faults under CT saturation, three distinct cases are examined.

The core idea is to employ WT-LSTM for image classification, wherein the image seemingly originates from a processed time series. Initially, a unique identifier representing one of the inrush current data is converted into a string, subsequently utilized to generate names for a PNG image and a MAT file. The model is tasked with generating and saving the PNG image, resizing it to dimensions

64×86 , and then saving the resized image. Subsequently, the image is classified using the model, with the classify function returning the class label. Upon checking the length of the label and extracting its first element, the classification result is compared. The image, along with its classification result, is displayed alongside a plot of relevant data. The simulation classifies the data, resulting in categorical data types. For instance, specific instances of IC detection with varying source angles are evident in Fig. 11a–c. This process is repeated for other scenarios, such as IDs for different fault types, where the algorithm efficiently detects internal

Table 6 Comparison of the efficiency and performance of the proposed method with recent SOTA researches

Method	Criteria				
	Signal type	Accuracy (%)	Cpu-time (Sec)	Considering CT saturation	Considering noisy data
Proposed method	External faults	96.2	0.191	✓	✓
	Inrush current	99.8	0.218		
	Internal faults	99.65	0.228		
	Inrush and internal fault	99.57	0.229		
[30, 31]	External faults	91.52	5.14	✓	×
	Inrush current	92.34	4.28		
	Internal faults	91.88	6.12		
	Inrush and internal fault	89.31	7.12		
[12, 33]	External faults	91.24	6.93	×	×
	Inrush current	91.39	5.29		
	Internal faults	87.39	8.79		
	Inrush and internal fault	—	—		

Bold values indicate that the proposed method exhibits superior accuracy in detecting all types of faults, while also requiring minimal computing time

faults within a quarter cycle, as depicted in Fig. 11d–f. The classifier accurately identifies the case as an “internal fault”. In cases of external faults with CT saturation, the spectral energy illustrated in Fig. 11g–i guides the model to classify the case as an “external fault”.

On the contrary, Table 6 presents a comparative analysis of our proposed method against recent advancements in research across various criteria, including CPU-time and accuracy, for different states of current signals. Upon examination of this table, it becomes evident that the proposed method exhibits superior accuracy in detecting all types of faults, while also requiring minimal computing time, thus approaching real-time performance. Specifically, the proposed method stands out for its ability to achieve high accuracy and efficiency across fault detection scenarios.

5 Conclusion

This study has developed a novel real-time end-to-end fault detection algorithm based on WT-LSTM, utilizing modified wavelet transform and LSTM to discern between IDs and ICs in PTs. The WT divides the differential current into multiple energy levels, while the LSTM is tasked with assigning useful characteristics to energy levels D1 to D5. The WT-LSTM architecture demonstrates faster identification of both IDs and ICs compared to other recent state-of-the-art deep-learning systems, owing to its expedited training process. Furthermore, the LSTM network effectively handles noisy inputs, such as signal distortion caused by capacitances and inductances of transmission lines, by filtering out irrelevant information, thus enhancing fault identification reliability.

Unlike traditional neural networks, LSTM networks excel in retaining information over long sequences, which is vital for identifying faults in power transformers. Their ability to maintain long-term memory through gated mechanisms equips them to handle various fault characteristics, including different types, magnitudes, and durations of inner faults in PTs. Moreover, the real-time capability of LSTM networks enables their deployment for real-time fault detection, facilitating timely alerts and enabling proactive maintenance strategies. Consequently, the stability and dependability of the power system can be augmented by the faster decision-making capabilities of WT-LSTM. Additionally, the suggested method requires minimal computational power, lacks the need for threshold values or vast amounts of training data. Simulation results indicate that WT-LSTM enhances classification accuracy, with 99.65% of all IDs reliably identified and distinguished from ICs using the proposed method. Finally, future research should also aim to include uncertainties with a high-precision model on measuring real-world data within an online fault detection framework. Such a method will enhance the dependability and accuracy of fault detection in real-time applications, thus improving the robustness and performance of the system. This proposed methodology will become improved from this standpoint because it can deal with these issues more efficiently and accurately thereby resulting into better and faster identification as well as minimization of faults in cases of anomalous variation in actual data.

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